

APSO MULTIMODAL AUTHENTICATION

Automated test and synthetic benchmark report | 8 October 2026

RESEARCH INTEGRITY: SYNTHETIC DATA ONLY - NOT HMOG OR PAPER RESULTS

1. Automated software tests

Command: PYTHONPATH=src python -m pytest -q
Result: 5 passed, 0 failed (0.75 seconds on the local test environment).
Covers: fusion rules, metric definitions, optimizer, bootstrap and SVM scoring.

2. Built-in synthetic demonstration (random seed 42)

CFA	Selected rule	Threshold	Held-out error
0.5	exp	1.5925	0.2480
1.0	sum	0.4796	0.2477
1.5	sum	0.5797	0.1776

3. Three-seed synthetic benchmark

400 generated attempts/seed, 2 modalities, 200 learning + 200 held out.
Swarm 20; 15 iterations; three seeds: 7, 21, 42.
Baseline: unweighted SUM, fixed 0.5 threshold (NOT methods [24] or [25]).

CFA	APSO mean E	Fixed sum mean E	APSO minus fixed
0.5	0.2350	0.2142	+0.0208
1.0	0.2384	0.2194	+0.0190
1.5	0.2132	0.2246	-0.0114

Interpretation: lower error is better. APSO is worse for CFA 0.5 and 1.0, but better for CFA 1.5, on average in these synthetic runs.

These values do NOT support general superiority or statistical significance.

Reproduce: pip install -e ".[dev]"; python -m pytest -q
Run: python experiments/synthetic_benchmark.py
Machine-readable outputs: reports/synthetic_benchmark_results.json

4. All nine synthetic benchmark runs

Seed	CFA	Rule	Threshold	APSO E	Fixed sum E
7	0.5	product	0.4328	0.1710	0.1609
7	1.0	product	0.4622	0.2047	0.1597
7	1.5	exp	1.7678	0.2216	0.1585
21	0.5	exp	1.6918	0.2861	0.2644
21	1.0	exp	1.7177	0.2628	0.2788
21	1.5	hamacher	0.3680	0.2404	0.2933
42	0.5	exp	1.5925	0.2480	0.2174
42	1.0	sum	0.4796	0.2477	0.2197
42	1.5	sum	0.5797	0.1776	0.2220

5. Important boundaries

- No HMOG or real participant recordings were used.
- No reference [24] or [25] implementations were evaluated.
- Synthetic score simulation does not reproduce published results.
- Original APSO parameters and feature extraction require validation.
- The positive/negative changes are descriptive, not significance tests.

6. Next steps for reproducibility

- Archive exact HMOG partitions and extracted scores.
- Confirm original APSO update equations and baseline methods.
- Reproduce 21 CFA levels on held-out real observations.
- Verify FAR/FRR, timings and paired per-user bootstrap intervals.